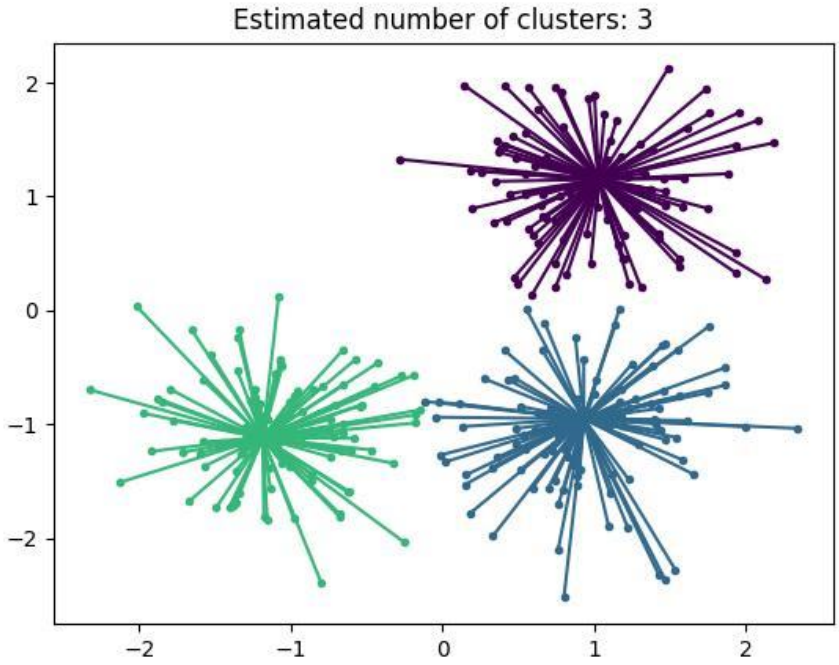
Clustering

Affinity Propagation:

Affinity Propagation is an unsupervised machine learning clustering algorithm that groups similar data points by identifying representative points called **exemplars**.

Unlike K-Means, Affinity Propagation does not require you to specify the number of clusters beforehand. Instead, it automatically determines the number of clusters based on similarities between data points and a parameter called *preference*.

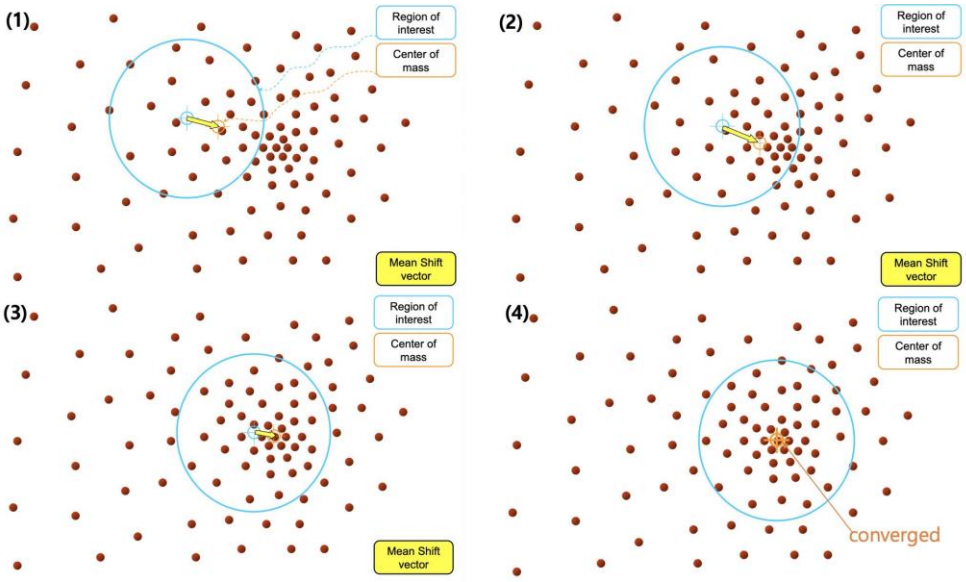


Mean Shift Clustering:

Mean Shift is an unsupervised machine learning clustering algorithm used to group similar data points based on the density of data points in a feature space.

It is a **centroid-based, non-parametric clustering algorithm** that does not require specifying the number of clusters in advance.

Clustering

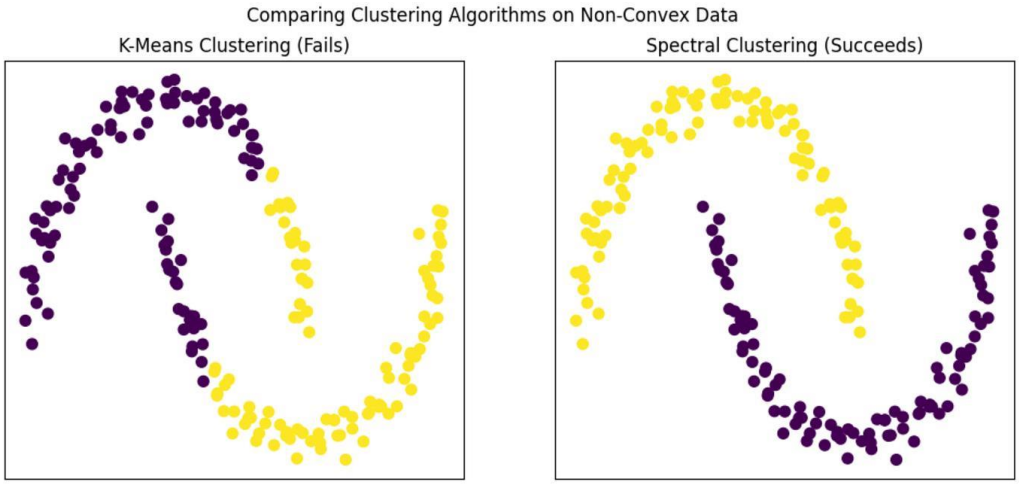


Spectral Clustering:

Spectral Clustering is an unsupervised machine learning algorithm used to group similar data points into clusters using the relationships (similarities) between data points.

It uses **graph theory and eigenvalues/eigenvectors of a similarity matrix** to identify clusters.

Unlike K-Means, Spectral Clustering can identify clusters with complex and non-spherical shapes.



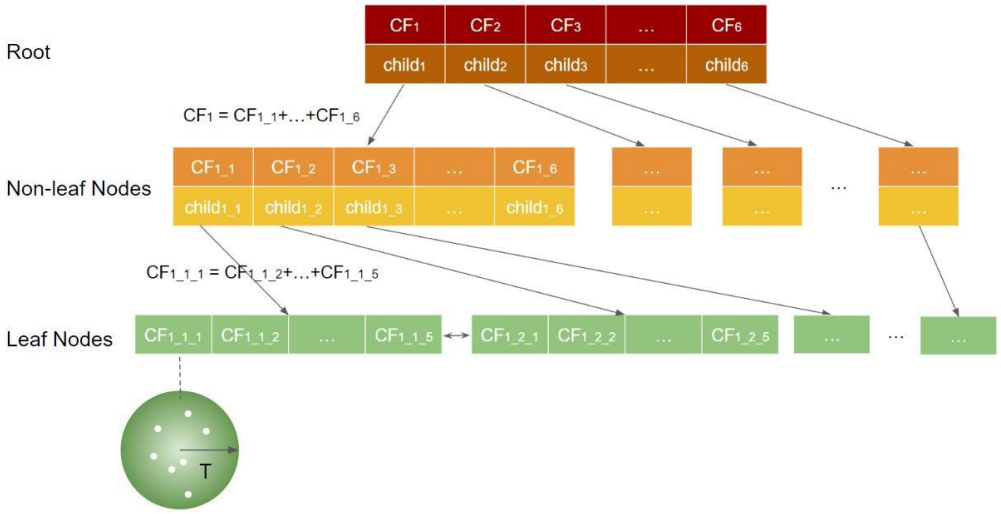
BIRCH Clustering:

BIRCH stands for Balanced Iterative Reducing and Clustering using Hierarchies.

BIRCH is an unsupervised machine learning clustering algorithm used to efficiently cluster large datasets, especially when the dataset contains many data points.

It uses a **tree structure called a CF Tree (Clustering Feature Tree)** to summarize data points and perform clustering efficiently.

Clustering

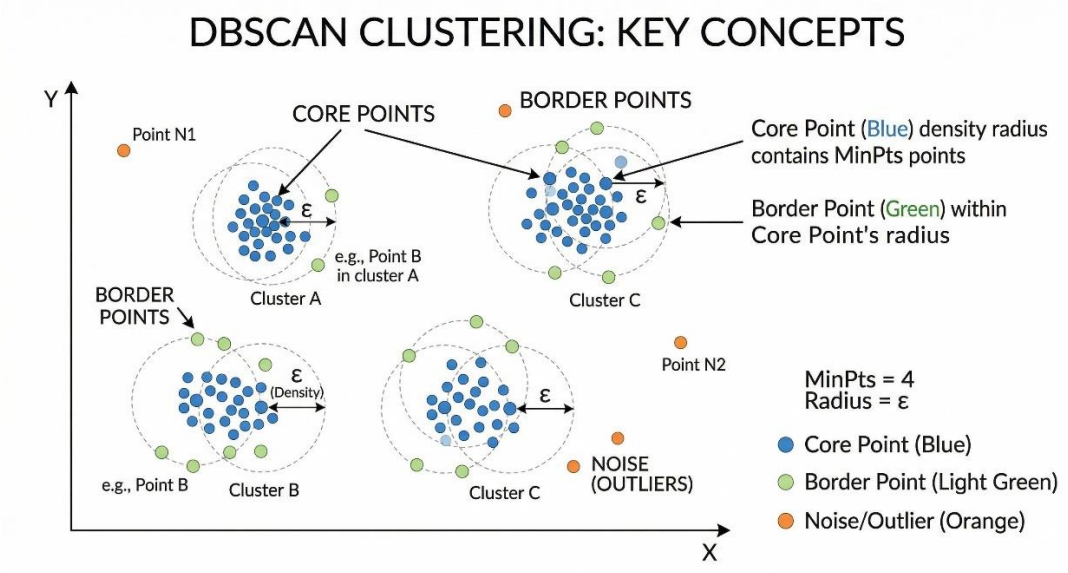


DBSCAN Clustering:

DBSCAN stands for Density-Based Spatial Clustering of Applications with Noise.

DBSCAN is an unsupervised machine learning clustering algorithm that groups data points based on their density.

It identifies clusters by finding regions where many data points are closely packed together. It also identifies isolated points as **noise or outliers**.



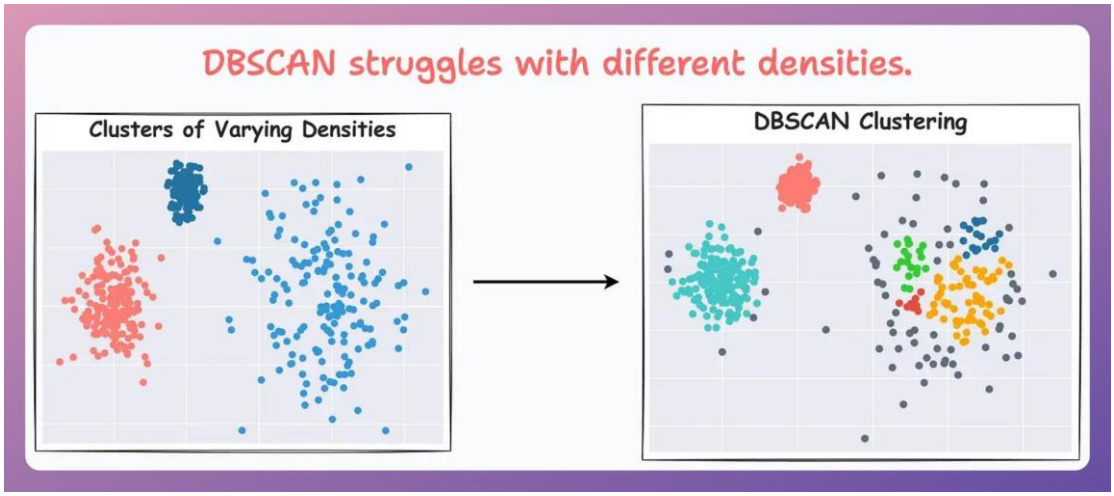
Clustering

HDBSCAN Clustering:

HDBSCAN stands for Hierarchical Density-Based Spatial Clustering of Applications with Noise.

HDBSCAN is an unsupervised machine learning clustering algorithm that extends DBSCAN by using hierarchical clustering and varying density levels.

It groups data points based on density and can identify clusters with different densities.



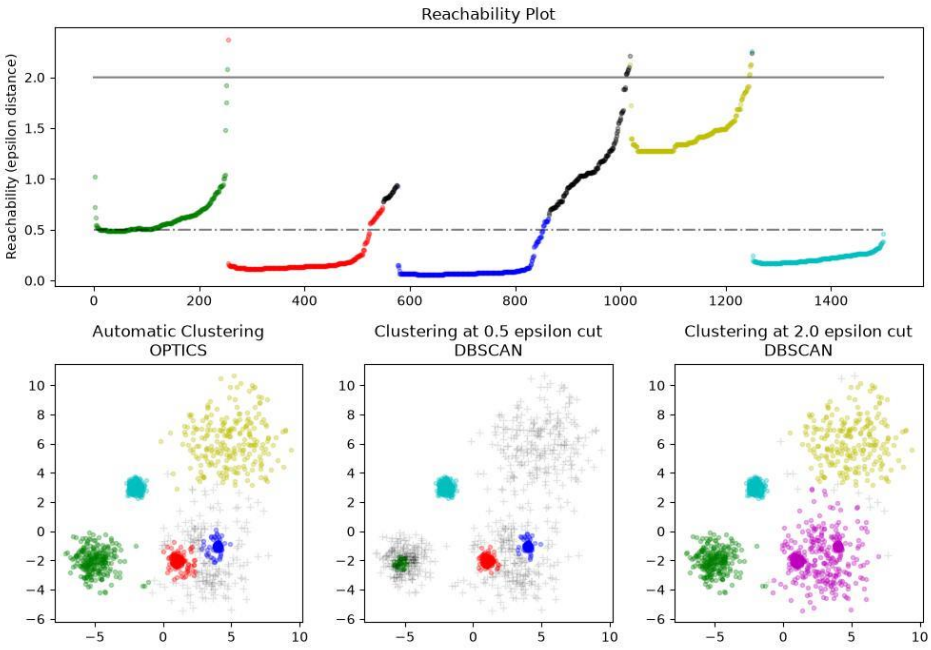
OPTICS Clustering:

OPTICS stands for Ordering Points To Identify the Clustering Structure.

OPTICS is an unsupervised machine learning clustering algorithm used to identify clusters in datasets containing different densities.

It is a density-based clustering algorithm similar to DBSCAN, but it can handle clusters with varying densities more effectively.

Clustering


Reachability Plot
Reachability (epsilon distance)
0.0
0.5
1.0
1.5
2.0
0
200
400
600
800
1000
1200
1400
Automatic Clustering
OPTICS
Clustering at 0.5 epsilon cut
DBSCAN
Clustering at 2.0 epsilon cut
DBSCAN